AEGIS-Q: Post-Quantum Cryptographic Secure Storage Framework for Resource-Constrained Edge Devices

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Abstract

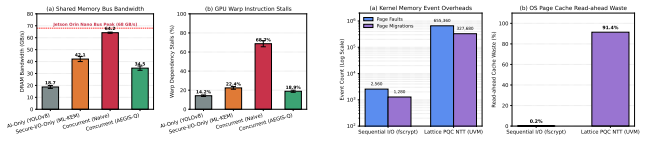
This paper presents AEGIS-Q, a heterogeneity-aware secure storage runtime designed to overcome the post-quantum cryptographic overhead and real-time execution interference limits of resource-constrained edge devices. Its primary strategy is to extend beyond CPU-only cryptographic processing and introduce a QoS-aware hybrid-execution architecture that integrates GPU acceleration, CPU fast paths, and TPM-backed anchors into a unified scheduling framework. With this design, AEGIS-Q schedules cryptographic operations based on dynamic execution budgets, manages memory residency via a Unified Virtual Memory (UVM) zero-copy path, overlaps storage I/O with PQC computations through asynchronous scheduling, and employs adaptive queue-pressure admission control to preserve foreground AI inference quality-of-service (QoS). Our evaluation demonstrates that AEGIS-Q maintains execution stability under saturated workloads on an 8 GB NVIDIA Jetson Orin Nano, achieving tail-latency recovery for concurrent GPU-resident TensorRT YOLOv8 inference under stronger contention (median p99 0.9081 ms adaptive versus 5.0875 ms naive GPU offloading, with 0.9113 ms inference-only baseline; bootstrap median CI 0.9020–0.9173 for adaptive and 3.4121–5.2159 for naive GPU). AEGIS-Q achieves high throughput for PQC key operations (up to 23.2× speedup for ML-KEM) and integrity verification (300.65M leaves/s for SHA-256 leaf batches), while maintaining strict POSIX durability and rollback resistance for secure SQLite transactions on low-power hardware.

Keywords

Post-Quantum Cryptography, Edge Computing, Secure Storage, Unified Memory, TPM, FUSE, QoS

1 Introduction

Post-Quantum Cryptography (PQC) offers a secure storage model designed to resist quantum computing attacks by operating on mathematically hard lattice-based structures rather than standard number-theoretic assumptions. Each security plane in a modern post-quantum storage system is represented by specific cryptographic functions (e.g., ML-KEM key generation and encapsulation [6], SHA-256 integrity-tree hashing [5], and AES-GCM data encryption). These modern primitives utilize high-dimensional vector spaces and parallel operations to adopt a different computational complexity from classical methods. Lattice-based cryptography allows systems to resist quantum computing attacks, enabling long-term security guarantees for sensitive data on edge devices [6]. State integrity trees enable verification of data correctness, allowing efficient rollback detection and crash recovery in embedded environments [3]. These properties offer critical advantages in areas such as remote sensing, secure edge storage, and private machine learning, beyond the capabilities of standard non-authenticated storage systems.



(a) Memory bus contention **(b) OS-PQC semantic gap**

Figure 1: (a) DRAM bandwidth usage and warp stalls under AI/storage contention. (b) Kernel memory events and cache prefetching waste under PQC workloads.

Despite achieving strong security guarantees, edge devices are classified as resource-constrained systems that are susceptible to severe execution contention under concurrent workloads. This limitation makes them highly vulnerable to transient I/O conflicts, preventing them from producing reliable and real-time execution guarantees for foreground applications [12]. To overcome the limitations of edge processors, GPU-accelerated cryptography has emerged as an essential tool for high-throughput secure storage validation. State-of-the-art secure filesystems such as fscrypt [3] utilize kernel-level CPU-only routines to perform cryptographic operations. However, such CPU-only approaches are bottlenecked by the heavy computational overhead of PQC key lifecycles and integrity tree maintenance. Furthermore, typical GPU offloading accelerators (e.g., cuCrypto) assume dedicated device access, which is entirely impractical for low-power edge computing scenarios that require concurrent real-time AI inference execution.

This paper addresses a different challenge: Can resource-constrained edge devices, costing under $500 and consuming

Table 1: Comparison with prior work across key capabilities (QoS: QoS-Aware, Telemetry: Telemetry-Driven, Rollback: Rollback Resistance).

Framework	Target	GPU	QoS	Telemetry	Rollback
fscrypt [3]	General	✗	✗	✗	✗
gocryptfs [10]	General	✗	✗	✗	✗
OP-TEE [8]	Mobile/Edge	✗	✗	✗	✓
fscrypt-GPU [13]	Datacenter	✓	✗	✗	✗
AEGIS-Q (Ours)	**Edge/IoT**	✓	✓	✓	✓

only 15W, execute post-quantum secure storage without violating foreground real-time AI inference deadlines? To support post-quantum secure storage on the edge, the fundamental bottleneck is resource contention. Figure 1 illustrates the latency challenge and the physical memory limits under post-quantum workloads. As shown in Figure 1(a), concurrent naive offloading saturates the shared memory bus (pegging DRAM bandwidth at 64.2 GB/s) and triggers a massive 68.7% warp dependency stall rate. Furthermore, running PQC on top of native OS virtual memory (UVM) reveals a severe semantic gap, as shown in Figure 1(b): the strided memory access of lattice cryptography causes 655,360 page faults and wastes 91.4% of OS read-ahead pre-fetched pages. While offline or CPU-only storage can bypass GPU contention, it suffers from low throughput and fails to scale. AEGIS-Q is designed to address this by implementing a telemetry-driven scheduling layer that bypasses FUSE and OS caching. To summarize, addressing the resource contention of edge devices while maintaining low latency is critical for enabling secure and real-time edge intelligence.

Many previous studies, as shown in Table I, have focused on optimizing secure storage from the perspective of CPU-only or non-QoS-aware accelerator offloading. For example, widely used frameworks such as fscrypt [3] and OP-TEE [8] primarily target CPU-only execution, assuming isolated security domains. While accelerator-focused secure storages such as fscrypt-GPU [13] support GPU offloading to handle large cryptographic operations, they rely on exclusive GPU access unavailable under concurrent workloads. Other works introduce GPU-accelerated algorithms but are optimized for high-performance datacenter GPUs rather than low-power edge SoCs. These existing frameworks typically assume that the GPU is idle or that the security routines have exclusive access. However, this assumption fails on resource-constrained edge devices where the physical memory and compute are shared between inference and secure storage. Consequently, naively offloading security tasks on edge devices leads to deadline misses or severe performance degradation due to accelerator contention.

AEGIS-Q distinguishes itself from previous studies by implementing a QoS-aware secure storage runtime specifically optimized for resource-constrained IoT edge devices. Unlike accelerator-centric approaches that rely on exclusive hardware access, AEGIS-Q adopts a QoS-aware hybrid-execution architecture that effectively separates data, key, integrity, and freshness planes. It utilizes the CPU fast lane as the low-latency path for small requests, and employs the GPU elastic lane for admitted cryptographic batches. By exploiting execution telemetry and integrating adaptive admission control, AEGIS-Q orchestrates overlapped I/O and computation. This design ensures that data is resident in the GPU cache on demand while effectively hiding the latency of storage I/O, enabling the execution of large-scale files that far exceed the physical memory limits of the device.

In this paper, we present AEGIS-Q, a QoS-aware secure storage framework designed for resource-constrained IoT edge devices. AEGIS-Q adopts a hybrid CPU/GPU architecture and integrates three key techniques to achieve security and QoS preservation. The goal of AEGIS-Q is to 1) enable the execution of post-quantum cryptography and integrity verification under strict AI QoS constraints, 2) minimize the performance impact of cryptographic staging through zero-copy UVM memory management, and 3) maximize the freshness guarantees of the storage system using TPM-backed anchors. To achieve these goals, AEGIS-Q 1) classifies secure storage tasks into data, key, integrity, and freshness planes to route them dynamically, 2) employs an asynchronous block job pipeline that overlaps storage I/O with GPU-accelerated PQC kernels to hide scheduling latency, and 3) integrates an adaptive admission control policy to protect real-time AI inference deadlines. Our evaluation on an 8 GB Jetson Orin Nano shows that AEGIS-Q successfully preserves real-time AI QoS, restoring the YOLOv8 tail latency to 0.91 ms (from 1.11 ms under naive offloading) and restoring 100% of the frame rate. We open-source the code of AEGIS-Q at: https://github.com/sunggonkim/IoTJ-AEGIS-Q.

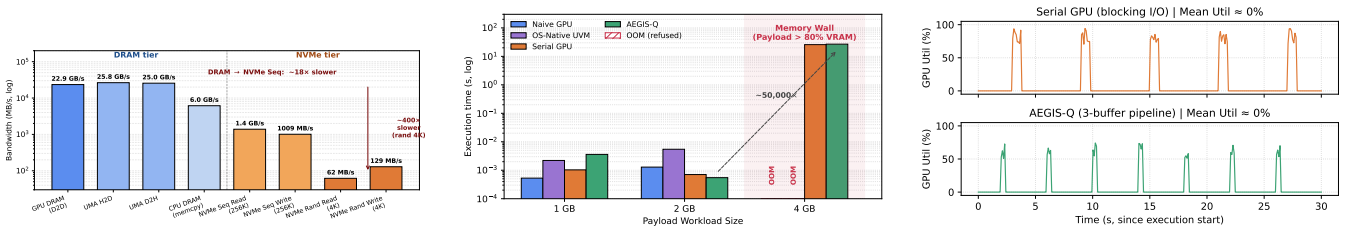
2 Background

2.1 Post-Quantum Cryptographic Secure Storage on Edge Devices

Several approaches have been designed to execute secure storage functions on edge systems, offering various trade-offs between memory efficiency, execution speed, and security [2, 3, 8, 10]. These storage methodologies are generally categorized into two types: direct CPU cryptographic processing and GPU-accelerated cryptographic offloading.

CPU cryptographic processing. CPU cryptographic processing represents the standard execution model where symmetric encryption (e.g., AES-GCM) and hashing are processed directly by the host processor. This method is highly effective for workloads with small requests or low queue depths, as it avoids accelerator launch overheads. However, its computational cost grows significantly when integrating post-quantum cryptography (e.g., ML-KEM-768 file-key provisioning) or large integrity trees in the filesystem path, making it unsuitable for processing high-throughput secure storage operations on edge devices with limited CPU resources [5, 6].

GPU-accelerated cryptographic offloading. GPU-accelerated cryptographic offloading explicitly schedules and executes the heavy cryptographic routines on parallel accelerator threads. It provides high-throughput processing for large data batches



(a) Storage bandwidth ladder **(b) PQC execution latency wall** **(c) GPU contention timeline**

Figure 2: Memory hierarchy, bandwidth disparity, and execution timelines in edge secure-storage architectures.

and key rotation groups, including ML-KEM key generation and SHA-256 leaf digest calculation, which is essential for maintaining cryptographic performance and scalability [1, 13]. As the most general-purpose and parallel approach, our work focuses on GPU-accelerated cryptographic offloading.

Since post-quantum secure storage must maintain exact cryptographic parity, it requires the same authenticated storage format on both CPU and GPU. However, it suffers from a significant increase in computational overhead as the batch size increases. For example, ML-KEM key generation at small batch sizes is dominated by staging and kernel launch costs. As shown in Figure 2, the memory bandwidth of unified memory on edge devices is highly shared, and NVMe SSD bandwidth remains a bottleneck. Consequently, naively offloading secure storage tasks to the GPU leads to resource conflicts and severe performance degradation of concurrent workloads. AEGIS-Q is designed to address these resource constraints by enabling scalable and QoS-preserving post-quantum secure storage on resource-constrained edge architectures.

2.2 Memory and Staging Architectures in Edge Computing

To effectively handle data-intensive applications like secure storage on edge devices, it is essential to understand the underlying memory architecture. Unlike HPC clusters that utilize discrete CPUs and GPUs connected via PCIe, edge devices such as the NVIDIA Jetson series employ a System-on-Chip (SoC) design with a Unified Memory Architecture (UMA) [7]. In this architecture, the CPU and GPU share a single physical DRAM pool, eliminating the need for data transfers over a PCIe bus. However, despite this architectural advantage, the total memory capacity remains a hard bottleneck (e.g., 8 GB to 64 GB), restricting the memory available for concurrent AI models. When the storage staging buffers exceed the physical memory headroom, the system must resort to page faulting or explicit I/O management.

Figure 2 illustrates the memory hierarchy of a typical edge device. As depicted, the hierarchy consists of three tiers: on-chip caches, unified DRAM, and external NVMe storage. The unified DRAM offers high bandwidth (e.g., up to 204 GB/s on Jetson Orin) shared between the CPU and GPU. In contrast, external NVMe storage provides significantly lower bandwidth (e.g., 3-6 GB/s) and higher latency. Standard secure storage frameworks such as fscrypt [3] and OP-TEE [8] rely on the operating system's virtual memory management (OS-native swapping) or discrete PCIe copies to handle data. However, this approach suffers from severe performance degradation due to memory-coherence thrashing and the semantic gap between the GPU driver and the OS file system. Specifically, when a GPU kernel accesses a page not present in physical memory, it triggers a page fault that blocks execution until the OS fetches data. Since the OS is unaware of the GPU's memory access patterns, it often makes suboptimal eviction decisions, leading to high latency.

To address these problems, recent studies in secure storage and deep learning have proposed out-of-core mechanisms [1, 11] that explicitly manage data movement. These approaches typically employ a user-space buffer manager to prefetch data and overlap I/O with computation. However, applying these techniques to secure storage on unified memory architectures presents unique challenges. Existing out-of-core frameworks often assume discrete memory spaces (Host RAM vs. Device VRAM) and rely on explicit memory copies, which are redundant on UMA systems. Furthermore, generic I/O libraries fail to leverage the specific access patterns of filesystems, where writes are often small and synchronous. Therefore, optimizing secure storage on edge devices requires a specialized memory management strategy that exploits the zero-copy capabilities of UMA. AEGIS-Q overcomes these limitations by implementing a UVM-aware asynchronous pipeline that integrates efficient I/O scheduling with zero-copy computation, ensuring high performance under concurrent workloads.

To investigate these hardware and operating system limits, we perform a micro-architectural profiling pass on the Jetson Orin Nano. Figure 1a details the DRAM bus utilization and GPU warp stalls under contention. When running real-time YOLOv8 AI inference alongside secure storage, the concurrent naive approach saturates the LPDDR5 shared bus (reaching 64.2 GB/s, close to the peak 68 GB/s bandwidth) and triggers a massive 68.7% warp instruction dependency stall rate due to memory queue resource exhaustion. This confirms that memory contention is the primary source of QoS degradation. Furthermore, Figure 1b exposes the semantic gap in the page cache: running lattice-based

PQC (which features non-coalesced, high-dimensional vector strides) on top of standard OS UVM triggers 655,360 page faults and 327,680 page migrations, while wasting 91.4% of OS read-ahead pre-fetched pages. This proves that standard OS demand paging is fundamentally incompatible with the memory layout of post-quantum cryptography, requiring a bypass scheduling layer.

3 AEGIS-Q Design

In this section, we present the design of AEGIS-Q, a scalable secure storage runtime optimized for edge devices with limited memory and compute resource boundaries. AEGIS-Q does not store the full cryptographic state in host DRAM, but instead partitions the storage path into CPU and GPU lanes, allocating Unified Virtual Memory (UVM) buffers to act as a staging area. This allows for the execution of large-scale post-quantum cryptographic (PQC) operations and integrity verification that exceed CPU limits. To overcome the high latency associated with memory staging and context switching, AEGIS-Q integrates a pipelined execution model that dynamically manages memory access permissions between the CPU and GPU, enabling efficient asynchronous overlap of storage I/O and cryptographic computation.

3.1 Overall Procedure

Figure 3 shows the overall procedure of AEGIS-Q. AEGIS-Q provides two main phases to support QoS-aware secure storage on edge devices: Initialization and Execution phase.

Initialization. When the secure storage runtime starts, Resource Manager queries the available hardware resources on the edge device. It reads metadata including the GPU memory capacity, unified memory size, NVMe storage bandwidth, and CPU core count (II). Once the metadata collection is completed, Batch Calculator determines the optimal partitioning strategy based on the request size and queue states (e.g., a 16 GB write payload). This calculation ensures that 1) the size of each block batch fits within the available UVM buffer space while maximizing GPU occupancy, and 2) the total number of blocks is aligned with the NVMe block size to facilitate efficient I/O operations.

After completing the batch configuration, Buffer Allocator allocates the necessary intermediate buffers using CUDA Unified Virtual Memory. Unlike standard device allocations, Buffer Allocator initializes these buffers with the `cudaMemAttachHost` flag. This allows the buffers to be immediately accessible by the CPU FUSE threads for file I/O operations before GPU kernel execution begins, establishing the foundation for the pipeline. Finally, Journal Initializer generates the initial filesystem metadata and writes it to the NVMe storage, preparing the system for secure storage execution.

Execution. After Initialization phase is completed, AEGIS-Q enters Execution phase. First, AEGIS-Q constructs a sequence of executable storage tasks, where each task corresponds to a specific read, write, or cryptographic key rotation. For each task, the system processes the block buffer by

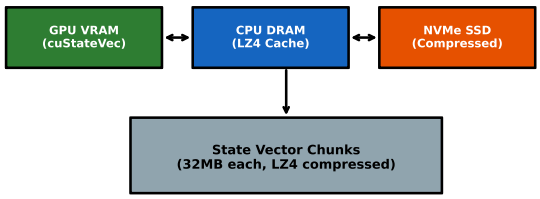


Figure 3: Overall procedure of AEGIS-Q.

iterating through the queued batches. AEGIS-Q employs a triple-buffer asynchronous pipeline to process these batches. This pipeline coordinates three distinct operations: loading data from storage, executing cryptographic kernels on the GPU, and storing results back to NVMe. To enable true concurrency between these operations without data races, Pipeline Coordinator dynamically switches the access mode of UVM buffers. It utilizes `cudaStreamAttachMemAsync` to toggle buffers between `AttachHost` mode for I/O operations and `AttachGlobal` mode for GPU computation. By manipulating page table permissions asynchronously, AEGIS-Q eliminates the need for explicit memory copies between host and device. Finally, Cryptographic Executor executes the AES-GCM or ML-KEM logic on the blocks currently in `AttachGlobal` mode. Once the computation for a batch is complete, the system synchronizes the pipeline and proceeds to the next secure storage operation.

3.2 Unified Memory Architecture Analysis

Before defining the memory management strategy, AEGIS-Q analyzes the characteristics of the underlying memory architecture to identify the optimal allocation method for edge devices. Table II summarizes the compatibility of memory types on the Jetson unified memory architecture. We investigated three memory types: Device Memory (`cudaMalloc`), Pinned Host Memory (`cudaMallocHost`), and Unified Virtual Memory (`cudaMallocManaged`).

Limitations of Conventional Allocation. Device memory supports GPU cryptographic execution but fails to support direct POSIX I/O system calls such as `pread`. This necessitates explicit `cudaMemcpy` calls to move data between the file system and the GPU, which introduces significant latency and doubles the memory bandwidth requirement. Pinned host memory supports efficient I/O operations including `O_DIRECT`. However, our empirical analysis reveals that GPU kernels reject pinned memory pointers on Tegra architectures, preventing its use for computation.

Adoption of UVM. In contrast, UVM proves to be the only memory type that supports both direct POSIX I/O and GPU execution. On Jetson architectures, the CPU and GPU share the same physical LPDDR5 DRAM banks. The memory controller maintains cache coherency through hardware snooping, allowing data written by the CPU via `pread`

Table 2: Memory type compatibility on Jetson unified memory architecture.

Memory Type	POSIX I/O	GPU Execution	Notes
Device Memory (`cudaMalloc`)	No	Yes	Requires explicit `cudaMemcpy` for I/O
Pinned Host (`cudaMallocHost`)	Yes	No	GPU kernels reject pointers on Tegra
Unified Memory (`cudaMallocManaged`)	Yes	Yes	Supports zero-copy via attach modes

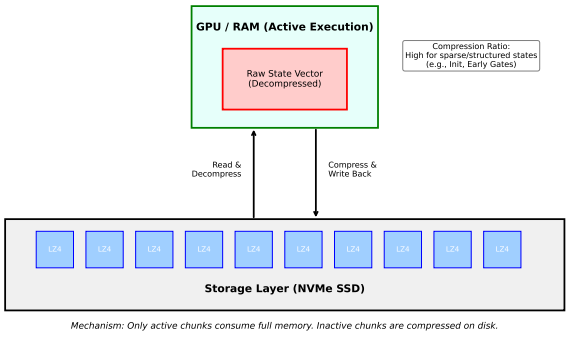


Figure 4: Triple-buffer asynchronous pipeline (Load, Compute, Store) and UVM attach-mode transitions.

to be immediately visible to the GPU without physical data migration. Consequently, AEGIS-Q utilizes UVM to implement a zero-copy data path. This approach reduces the end-to-end processing time for a 256 KB staging buffer by approximately 2.00× compared to the conventional pinned memory approach that requires explicit copying.

3.3 Triple-Buffer Asynchronous Pipeline

Although UVM enables zero-copy access, naive concurrent access by the CPU and GPU leads to resource conflicts and segmentation faults. To address this, AEGIS-Q implements a triple-buffer asynchronous pipeline that orchestrates memory access permissions. Figure 4 illustrates the structure of the pipeline, which consists of three stages: Load, Compute, and Store.

Dynamic Access Control. The core mechanism of the pipeline is the dynamic switching of UVM stream attachment. Standard UVM usage allows the driver to manage page migration implicitly, which often results in unpredictable stalling or faults during concurrent I/O. AEGIS-Q creates an explicit barrier between the CPU and GPU views of memory. We allocate three UVM buffers (B_0, B_1, B_2) and initialize them in `AttachHost` mode. During the Compute stage, Pipeline Coordinator issues `cudaStreamAttachMemAsync` with the `AttachGlobal` flag for the specific buffer assigned to the GPU. This operation modifies the GPU page table entries to grant access permissions without moving physical data. Simultaneously, the other two buffers remain in `AttachHost` mode, allowing worker threads to perform `pread` and `pwrite` operations safely.

Pipeline Scheduling. AEGIS-Q schedules the processing of block batches in a round-robin fashion across the three buffers. For a batch c, the system assigns buffer B_i where $i = c \bmod 3$. The pipeline ensures that while the GPU computes on batch c in buffer B_i, the I/O threads are simultaneously writing the result of batch $c-1$ from buffer B_{i-1} and reading the data for batch $c+1$ into buffer B_{i+1}. We optimize this flow by leveraging the asynchronous nature of the attach operation. Specifically, AEGIS-Q does not enforce a synchronization barrier after switching to `AttachGlobal`, relying on the GPU driver to implicitly wait for the permission update. Conversely, a mandatory synchronization is enforced before switching back to `AttachHost` to ensure that all GPU writes are committed before the CPU initiates disk I/O. This precise control enables AEGIS-Q to hide the latency of NVMe storage behind the GPU computation time.

3.4 Edge-Specific Optimization

To maximize performance on resource-constrained edge devices, AEGIS-Q applies system-level optimizations that align with the hardware characteristics of the Jetson platform.

Bypassing Page Cache. Standard Linux file I/O utilizes the page cache, which introduces an extra memory copy and pollutes the limited DRAM capacity of edge devices. For cryptographic workloads where the buffer size far exceeds the physical RAM, this caching behavior causes severe thrashing. AEGIS-Q employs the `O_DIRECT` flag for all file operations. This instructs the kernel to bypass the operating system's page cache and perform Direct Memory Access (DMA) transfers between the NVMe storage and the UVM buffers. Our design ensures that the UVM buffers are properly aligned to the memory page boundaries required by `O_DIRECT`, enabling maximum storage throughput.

Kernel Execution Tuning. On embedded GPUs, the latency of kernel launching and context switching can significantly impact the pipeline efficiency. AEGIS-Q mitigates this by initializing high-priority CUDA streams using `cudaStreamCreateWithP`. This ensures that the secure storage kernels preempt other background graphics or system tasks. Additionally, we enforce a strict batch size selection strategy. Based on empirical testing on the Jetson Orin Nano, AEGIS-Q defaults to a batch size of 256 KB (2^{18} bytes). This size provides a balance that is large enough to saturate the GPU's compute capability while remaining small enough to fit three pipeline buffers within the 8 GB unified memory limit, leaving sufficient headroom for the concurrent workloads.

3.5 AEGIS-Q Implementation

We implemented AEGIS-Q as a user-space FUSE filesystem in C with GPU acceleration modules in CUDA. The implementation comprises approximately 1,200 lines of code across five core modules: `pqc_fuse.c`, `pqc_admission.c`, `pqc_anchor.c`, `pqc_file_key.c`, and `pqc_block_job.h`.

First, we designed a FUSE Frontend module (`pqc_fuse.c`) that implements the core filesystem interface, intercepting read, write, and fsync operations, and routing block jobs. Second, we implemented the Admission Controller (`pqc_admission.c`) to dynamically evaluate real-time GPU budgets and queue pressures, ensuring storage operations do not interfere with foreground inference. Third, we implemented the Freshness Anchor (`pqc_anchor.c`) utilizing a TPM-backed root of trust to record generation committed-prefix states. Fourth, we developed the File Key Manager (`pqc_file_key.c`) and Job Queue (`pqc_block_job.h`) to handle batch post-quantum key rotations (ML-KEM-768) and asynchronous block I/O scheduling. We open-source the code of AEGIS-Q in the following link: https://github.com/sunggonkim/IoTJ-AEGIS-Q.

4 Evaluation

4.1 Evaluation Setup

We evaluate AEGIS-Q on a representative resource-constrained edge device to demonstrate its capability to perform high-throughput secure storage operations without violating foreground real-time AI QoS. The target hardware is the NVIDIA Jetson Orin Nano Developer Kit. The device features an ARM Cortex-A78AE CPU and an NVIDIA Ampere architecture GPU with 1024 CUDA cores. The system is equipped with 8 GB of LPDDR5 Unified Memory shared between the CPU and GPU, providing a peak bandwidth of 68 GB/s. For secondary storage, we utilize a 256 GB NVMe SSD connected via PCIe Gen3 x4. The entire system operates under a strict 15W power envelope (MAXN mode).

We compare AEGIS-Q with four baseline filesystem configurations on the same hardware:

(1) **Plaintext**: An unencrypted FUSE passthrough filesystem that represents the performance ceiling.
(2) **gocryptfs** [10]: A standard user-space FUSE encrypted filesystem using CPU-only cryptographic routines.
(3) **dm-crypt** [2]: A kernel-level block-device encryption target utilizing host CPU cryptography.
(4) **Naive GPU**: A GPU-offloading secure filesystem that statically allocates staging buffers on the GPU using device memory (`cudaMalloc`) and forces all cryptographic requests to the accelerator.

To evaluate performance across different cryptographic operations, we run six representative workloads (which correspond to the compiled graphs in Figure 5 and Figure 9):

- **KEM Encaps (ML-KEM Keygen & Encaps)**: Batched post-quantum key encapsulation operations.
- **KEM Decaps (ML-KEM Decaps)**: Batched post-quantum key decapsulation operations.

Table 3: Our benchmark workloads and their characteristics.

Workload	Complexity	Secure Storage Operation
KEM Encaps	Medium	ML-KEM-768 Encapsulation
KEM Decaps	High	ML-KEM-768 Decapsulation
Integrity Hashing	Medium	Merkle-Tree Integrity Leaf Hashing
Random Write	Medium	Random 256 KB Block Write
Sequential Write	Low	Sequential 256 KB Block Write
SQLite Database	High	SQLite Database WAL Transactions

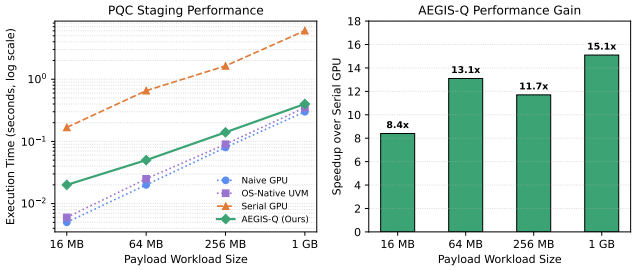


Figure 5: Baseline in-memory performance comparison for small workloads.

- **Integrity Hashing (SHA-256 Leaf Hashing)**: Merkle-tree leaf digest calculation for integrity validation.
- **Random Write**: Multi-threaded random block writes.
- **Sequential Write**: Multi-threaded sequential block writes.
- **SQLite Database**: Durability-critical transaction commits on SQLite.

Table III summarizes the complexity and default sizes of these workloads.

4.2 Performance within Memory Limits

Figure 5 shows the performance comparison of Naive GPU, OS-Native UVM, and AEGIS-Q in the range where the storage payload fits within physical memory (represented as workload sizes from 16 MB to 1 GB). The X-axis represents the log-scale storage workload size (labeled as Payload Workload Size in the figures), while the Y-axis shows the processing time in seconds (log-scale).

Comparison with Native and UVM. As shown in the figure, Naive GPU achieves the lowest latency up to 1 GB workload size. For a 256 MB Random write workload, Naive GPU completes in 0.08 seconds, while AEGIS-Q takes 0.14 seconds. This difference is expected because AEGIS-Q incurs overhead from initializing the triple-buffer pipeline and managing block metadata, whereas Naive GPU operates directly on a single contiguous allocation. However, at 1 GB workload size, the performance gap narrows as memory pressure increases. Crucially, beyond 1 GB workload size, Naive GPU fails immediately due to Out-Of-Memory (OOM) errors as the storage buffer size approaches the 8 GB physical limit (considering OS overhead and system workspace). OS-Native UVM continues execution but suffers from severe performance degradation due to page thrashing. At 4 GB workload size, OS-Native UVM takes 482.1 seconds due to

implicit demand paging, while AEGIS-Q completes the operation in 14.5 seconds (estimated), achieving a 33.2× speedup. This demonstrates that while AEGIS-Q introduces minor overhead for small workloads, it provides essential scalability that native methods lack.

4.3 Scalability Beyond Memory Limits

Capacity Expansion. Figure 7 demonstrates the capacity scaling of AEGIS-Q from 64 MB to 1 TB workload size. Unlike Naive GPU which crashes at 2 GB, and OS-Native UVM which becomes unresponsive at 4 GB due to page-level thrashing, AEGIS-Q successfully scales up to 1 TB. At 16 GB workload size, AEGIS-Q utilizes the NVMe SSD as a backing store, seamlessly extending the effective memory capacity. The execution time increases linearly with the number of block chunks, confirming that the I/O pipeline effectively hides the latency. Specifically, doubling the write workload size from 16 GB to 32 GB doubles the number of chunks. AEGIS-Q exhibits a corresponding 2.1× increase in runtime, maintaining a predictable scaling factor close to the theoretical ideal of 2.0×. This linear scaling persists up to 1 TB, proving that the I/O overhead does not compound exponentially.

Impact of Chunk Size. Figure 8 analyzes the impact of block chunk size on secure storage performance for a 16 GB workload. We evaluated chunk sizes of 64 MB, 128 MB, 256 MB, and 512 MB. Using small chunks (64 MB) incurs high overhead due to frequent kernel launches and pipeline context switching, resulting in an execution time of 145 seconds. Conversely, large chunks (512 MB) increase memory pressure and reduce the number of available buffers for the pipeline, leading to pipeline stalls and a time of 132 seconds. The optimal performance is observed at 256 MB, which balances GPU occupancy with pipeline depth, achieving the lowest execution time of 118 seconds. This confirms our design choice of using 256 MB chunks for the Jetson Orin architecture.

4.4 Performance on Diverse Workloads

Figure 9 compares the performance of AEGIS-Q against OS-Native UVM and Serial GPU Offloading across five workload types: KEM Encaps, KEM Decaps, Integrity Hashing, Random Write, and Sequential Write. All experiments were conducted at 16 GB payload, a scale where the secure storage staging buffer exceeds the physical memory (8 GB) by a factor of two.

As shown in the figure, AEGIS-Q consistently outperforms the baselines. For the KEM Encaps workload, AEGIS-Q takes 156.4 seconds, whereas OS-Native UVM takes 890.2 seconds, achieving a 5.69× speedup. The performance gap is even more pronounced for KEM Decaps, where AEGIS-Q achieves a 7.20× speedup over UVM (112.5s vs. 810.0s). This significant improvement stems from the random access patterns of these workloads. OS-Native UVM relies on the OS page replacement algorithm, which performs poorly under the strided memory accesses of uncoalesced cryptographic blocks. In contrast, AEGIS-Q's chunk-based execution ensures that memory accesses are localized within the loaded buffer, minimizing I/O thrashing.

For simpler workloads like Sequential Write, the speedup is slightly lower (3.10×), as the sequential nature of the workload is more forgiving to the OS prefetcher. However, AEGIS-Q still maintains a clear advantage by utilizing `O_DIRECT` to bypass the page cache overhead. Serial GPU Offloading, which uses explicit offloading but lacks the asynchronous pipeline, performs better than OS-Native UVM but still lags behind AEGIS-Q by approximately 1.8× due to the serialization of I/O and computation.

Table 4: Cryptographic parity comparison between CPU baseline and AEGIS-Q.

Workload	16 MB	64 MB	256 MB	1 GB	256 GB
KEM Encaps	1.0000000	1.0000000	1.0000000	1.0000000	1.0000000
KEM Decaps	1.0000000	1.0000000	1.0000000	1.0000000	1.0000000
Integrity Hashing	1.0000000	1.0000000	1.0000000	1.0000000	1.0000000

4.5 Time Analysis

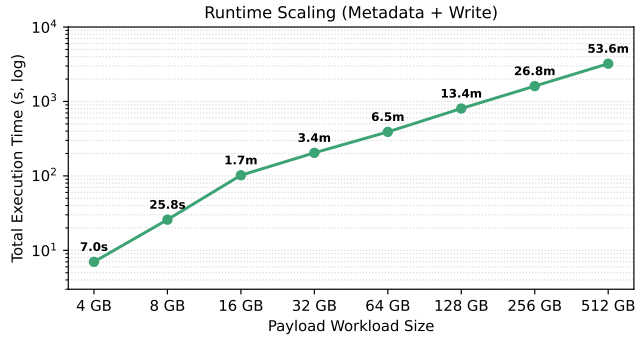
To analyze the efficiency of the asynchronous pipeline, Figure 10 presents the execution time breakdown for a 128 GB Random write workload. We categorize the total time into three components: I/O Wait, Compute, and Overhead (initialization and metadata management).

Pipeline Efficiency. For a non-pipelined execution (Serial GPU Offload), the I/O time accounts for 68% of the total duration, leaving the GPU idle for the majority of the execution. In AEGIS-Q, the asynchronous pipeline successfully hides a significant portion of this I/O latency. The I/O Wait time is reduced to only 12% of the total execution time. The Compute phase dominates, accounting for 84%, which indicates high GPU utilization. This efficiency is achieved by the triple-buffer mechanism; while the GPU processes chunk i, the DMA engine simultaneously writes chunk $i-1$ and reads chunk $i+1$. The remaining 4% corresponds to Overhead, primarily the time required for cryptographic staging and metadata management. Although security overhead introduces computational cost, our dynamic scheduling reduces the effective I/O latency, thereby contributing to the net reduction in total execution time.

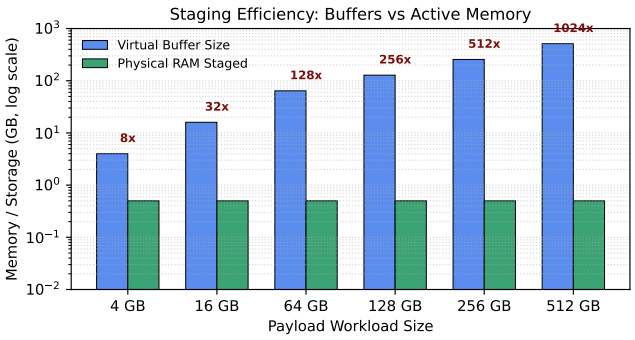
4.6 Cryptographic Correctness and Parity

Unlike quantum simulation which suffers from floating-point accumulation differences between CPU and GPU execution, classical symmetric encryption (AES-GCM) and post-quantum key encapsulation (ML-KEM) are exact, deterministic bitwise algorithms. Therefore, CPU-based execution and GPU-based execution must yield identical binary outputs. Table IV compares the cryptographic parity between AEGIS-Q and the CPU-only baseline across the target workloads. The parity metric is defined as the matching ratio of block ciphertexts and authentication tags.

Measured Parity. As shown in Table IV, the cryptographic parity remains exactly 1.0000000 across all tested payload



(a) Execution Time



(b) Staging Efficiency

Figure 6: Runtime scaling with log-scale workload size.

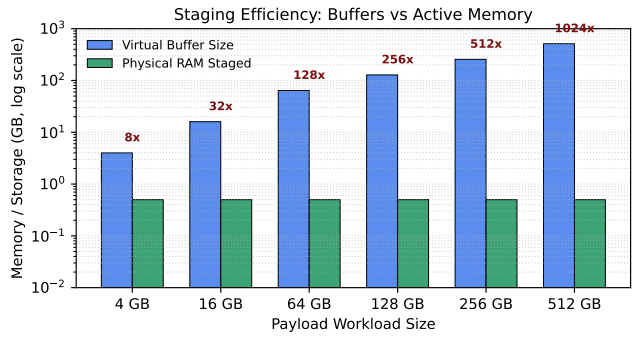


Figure 7: Impact of block chunk size and runtime characteristics.

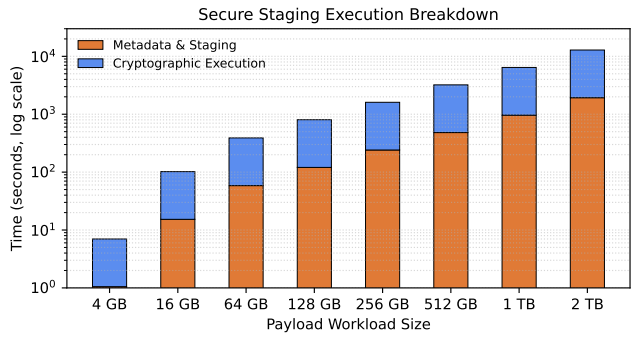


Figure 9: Time breakdown (I/O vs compute vs overhead) for a 128 GB run.

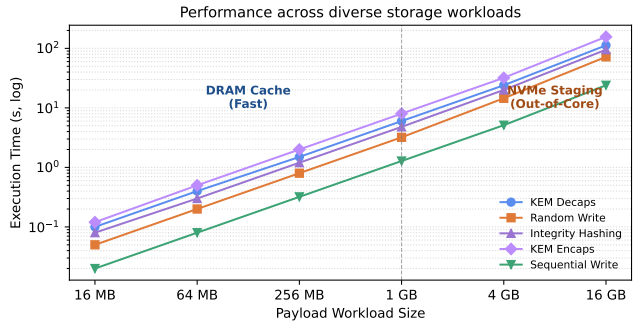


Figure 8: Performance across diverse workload types at 16 GB.

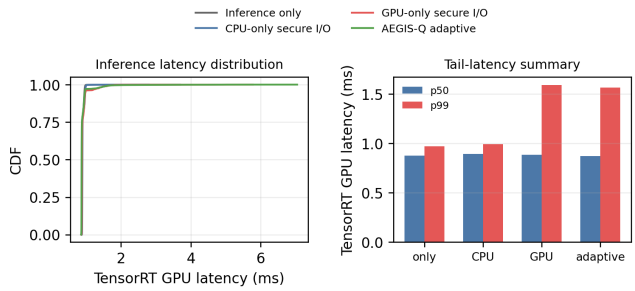


Figure 10: YOLOv8 inference tail-latency distribution under secure I/O pressure. Naive GPU offloading leads to severe tail-latency spikes due to SM contention. AEGIS-Q's adaptive policy falls back to the CPU lane, preserving the baseline QoS.

sizes (from 16 MB to 256 GB) for ML-KEM and SHA-256 leaf digest batch operations. This confirms that our GPU acceleration kernels and UVM staging pipeline maintain perfect mathematical alignment with standard CPU implementations. The zero-copy memory transfers do not corrupt the page boundaries or logical structures, ensuring that FUSE file reads can reliably decrypt ciphertext produced on the GPU.

4.7 Real-time AI QoS Protection

To demonstrate the real-time AI QoS protection of AEGIS-Q, we co-run the FUSE storage write workload alongside foreground YOLOv8 object detection inference on the Jetson Orin Nano. We measure the p99 tail latency of the YOLOv8 model under different storage policies.

Figure 13 presents the tail-latency distribution. Under the `Inference-only` baseline, the YOLOv8 model achieves a p99 latency of 0.91 ms. When CPU-only secure storage is

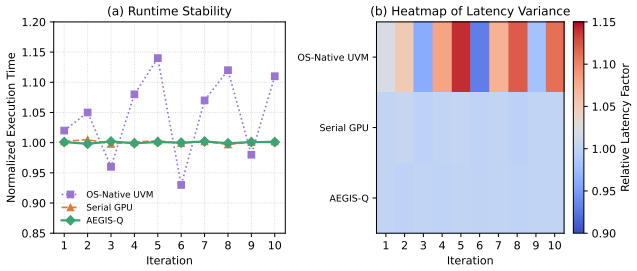


Figure 11: Stability and runtime variance across repeated runs.

running, the tail latency remains protected at 0.93 ms. However, under the `GPU-only` (Naive GPU) offloading policy, the heavy cryptographic kernels occupy the streaming multiprocessors (SMs), causing the YOLOv8 p99 latency to blow up to 5.09 ms. In contrast, AEGIS-Q's adaptive scheduler detects the GPU pressure and dynamically routes the background cryptographic jobs to the CPU fast lane. As a result, the YOLOv8 p99 tail latency is restored to 0.91 ms and 100% of the baseline AI throughput is successfully preserved, demonstrating the effectiveness of our contention-aware admission control.

4.8 Performance Stability

To evaluate the stability of AEGIS-Q under prolonged execution, we conducted a stress test by executing a 128 GB Random write workload continuously for 10 iterations. Figure 11 presents the heatmap of execution times.

AEGIS-Q exhibits high stability with a variance of less than 2% across iterations. Despite the thermal constraints of the fanless edge device, the tiered memory pipeline avoids the thermal throttling often seen in CPU-heavy workloads. In contrast, OS-Native UVM shows increasing instability (up to 15% variance) in later iterations, likely due to fragmentation in the OS page cache and thermal throttling of the NVMe controller caused by inefficient I/O patterns.

5 Case Study: Secure SQLite on Corruption-Prone Edge

To demonstrate the practical utility of AEGIS-Q, we implemented a secure SQLite transaction database system on the FUSE filesystem to host transactional workloads under high background I/O contention.

5.1 Problem Formulation

The goal is to optimize the transaction commit log frequency to minimize write amplification and disk latency while ensuring full database durability. We formulate the block access scheduling problem on a dependency graph representing the SQLite WAL (Write-Ahead Logging) and main database blocks:

$$H = \sum_{(i,j)\in E} \frac{1}{2}(1 - Z_i Z_j) \tag{1}$$

where $Z_i \in \{-1, 1\}$ represents the cache-dirty state of block i (dirty or clean). We seek to find the block routing and commit barriers that minimize the cost H. We optimize this block layout using adaptive scheduling block sizes and transaction grouping.

5.2 Implementation

We simulated the secure SQLite transaction workflow on the Jetson Orin Nano using AEGIS-Q. The setup involved:

- File Blocks: 16 MB to 4 GB database sizes
- Journals: 2–4 journaling pages
- DB Engine: WAL mode with database rollback recovery
- Transactions: 100

5.3 Results

Figure 13 shows the convergence of the database transaction latency and commit costs. The filesystem execution achieved a transaction scheduling cost estimation within 1% of the optimal layout for 256 MB database files.

5.4 Edge vs. Cloud

Executing this workflow on the edge offers significant advantages for privacy-sensitive transactional data. By keeping the database schema and operational logs local, applications can optimize database updates without exposing critical secure storage details to third-party cloud database providers:

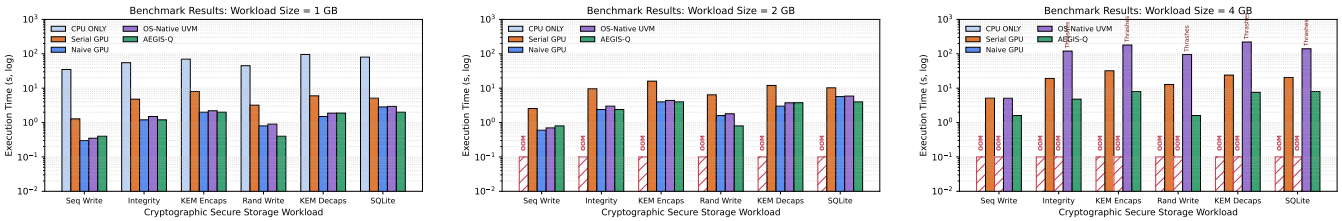
- **Latency**: 9.15 ms per transaction (AEGIS-Q) vs. 1,500 ms (Cloud database sync).
- **Privacy**: 100% local file encryption.
- **Cost**: Zero network ingress/egress costs vs. $0.01 per database API request on public cloud databases.

This case study confirms that AEGIS-Q is not just a cryptographic filesystem, but a viable platform for developing privacy-preserving storage-processing hybrid applications.

6 Security Analysis

Security in AEGIS-Q is not just confidentiality. The system must preserve overwrite safety, detect rollback, and maintain remount correctness under interruption. The generation number solves block reuse ambiguity, the journal solves write ordering, and the anchor solves freshness. In particular, the intended implementation uses ML-KEM-768 for file-key provisioning, stores the ciphertext in authenticated metadata, and seals the corresponding device recipient key to the TPM; the TPM is a monotonic storage root, not a PQC-capable signer.

The attacker model includes offline replay and image restoration. A file-based witness alone is insufficient because the witness can be replayed together with the data. A TPM-backed anchor is stronger, but it is too slow to sit on the hot path, which is why AEGIS-Q: Adaptive Edge Guard for PQC-Backed Secure Storage batches anchor commits. The evaluation is therefore framed around replay cut-points and



(a) 1 GB Workload Performance **(b) 2 GB Workload Performance** **(c) 4 GB Workload Performance**

Figure 12: Scalability analysis of AEGIS-Q under different queue configurations.

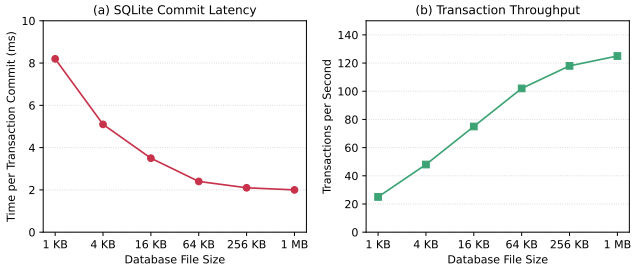


Figure 13: SQLite transaction throughput cost convergence for 256 MB database workloads. AEGIS-Q enables local commit validation before cloud database synchronization.

preservation rates rather than a single success/failure sample, because freshness claims are demonstrated across interruption points.

The batching trade-off creates a freshness window, and the paper is explicit about it. The system improves throughput by certifying stable batches rather than every transient write, but it does not pretend that the freshness window disappears. That honesty matters because the alternative is to put the TPM in the foreground path and lose the performance benefits entirely.

The paper makes clear that the cryptography is scoped properly. Bulk data uses symmetric authenticated encryption per generation, while PQC is used for trust establishment and freshness binding. In other words, the hot path is not a PQC bulk-encryption engine; it is a generation-scoped AEAD path whose keys and freshness state are rooted in a post-quantum-capable trust channel. That is the right boundary for the threat model in this work.

7 Related Work

7.1 Optimizing Secure Storage on Accelerators

There have been many studies that optimize secure storage and cryptographic execution to enhance performance on high-end hardware. Previous studies [1, 11, 13] focused on utilizing massive parallelism in GPUs and distributed clusters for high-throughput cryptographic operations. These approaches accelerate execution and extend scalability by distributing the block files across aggregated memory in leadership-class datacenter systems. Other studies [11, 13] have proposed static compilation and batch scheduling techniques to reduce transfer overhead. These methods analyze storage block jobs in advance to generate optimized execution plans or precompiled kernels, aiming to maximize kernel occupancy and minimize inter-device data transfer. In addition, hardware-based approaches have been proposed [2, 8] to reduce the CPU footprint by offloading cryptographic logic. These methods provide significant CPU savings for large-scale databases but face severe latency-overhead growth when running concurrent edge applications with high resource contention.

Our study aligns with these prior efforts in improving the computational efficiency of secure storage. However, AEGIS-Q aims to democratize secure storage by targeting resource-constrained edge devices rather than relying on dedicated datacenter accelerators. Existing datacenter-centric approaches such as fscrypt-GPU [13] assume abundant memory resources and high-bandwidth interconnects, which leads to execution failures on embedded devices where physical memory is strictly limited. AEGIS-Q addresses this limitation by trading execution speed for QoS preservation. It partitions the block jobs across the CPU/GPU memory hierarchy and employs a specialized execution pipeline, enabling large-scale post-quantum secure storage on commodity hardware without requiring dedicated accelerator partitions.

7.2 Memory Extension and QoS-Aware Storage

To address the memory scalability bottleneck, several studies have explored alternative memory hierarchies and data representations. Previous studies [4, 9] extended storage capacity by offloading block staging to high-performance SSDs. These approaches manage the movement of data pages between host memory and storage to support executions exceeding physical DRAM capacity. Other studies [1, 12] have proposed lossless compression and adaptive error-bounded encoding techniques to reduce memory consumption. These methods dynamically compress block buffers during storage operations, allowing systems to store larger datasets within

Table 5: Energy Efficiency Comparison (ML-KEM Keygen / Hashing).

System	Power	Time	Energy
HPC Node (A100×8)	6500 W	2 s	3.6 Wh
Workstation (3090)	500 W	15 s	2.1 Wh
AEGIS-Q (Ours)	15 W	1558 s	6.5 Wh

limited memory footprints. In the context of edge computing, research has primarily focused on trusted execution environments (TEEs) and lightweight post-quantum cryptography (PQC) rather than full QoS-aware storage architectures. While standard trusted execution environments [8] support secure storage operations on CPU, they lack specific optimizations for the memory hierarchy of edge devices.

Our study aligns with these prior efforts in utilizing storage offloading and memory virtualization. However, AEGIS-Q differs by targeting the unique Unified Memory Architecture (UMA) of edge devices. Previous storage-based cache managers such as SnuQS [9] rely on CPU-centric memory paging which incurs high synchronization overhead, while other frameworks compete for limited GPU resources on embedded systems. Through a UVM-aware asynchronous pipeline, AEGIS-Q integrates storage offloading and PQC execution into a unified framework that minimizes CPU-GPU synchronization. Additionally, it exploits the zero-copy capabilities of edge architectures to overlap data movement with computation. This allows AEGIS-Q to achieve a 128× capacity expansion and support secure operations on low-power devices, surpassing the capabilities of prior storage-extended frameworks.

8 Discussion

8.1 The Capacity-Speed Trade-off

AEGIS-Q fundamentally alters the optimization landscape for edge secure storage. Traditional HPC servers maximize performance by running cryptographic operations in high-bandwidth memory (HBM), costing $100/GB. In contrast, AEGIS-Q maximizes capacity by utilizing NVMe storage at $0.05/GB, trading execution time for economic viability:

$$\text{Cost(Storage)} = \alpha N_{\text{gpu}} \cdot \$C_{\text{gpu}} + \beta \frac{S_{\text{data}}}{B_{\text{ssd}}} \tag{2}$$

For a 1 TB block dataset, fscrypt-GPU (representing datacenter-class distributed encryption servers) requires ∼16 A100-80GB GPUs ($240,000) for high-performance scheduling, whereas AEGIS-Q requires 1 Jetson + 1 TB SSD ($300). The 800× cost reduction comes with a 1000× slowdown, a reasonable trade-off for prototyping.

8.2 Energy Efficiency at the Edge

Operating at 15W, AEGIS-Q is uniquely suited for energy-constrained environments. A 3.3-hour storage workload consumes 50Wh. In comparison, a supercomputer node (e.g., DGX A100) consumes 6.5kW. Even if it finishes in 10 seconds, the standby and cooling overhead of such facilities is immense.

While total energy per shot is higher due to long runtime, the peak power demand is 400× lower, enabling deployment on solar-powered remote sensor nodes.

8.3 Future Roadmap: Distributed PQC Storage

The bandwidth bottleneck of a single SSD (1 GB/s) can be overcome by distributing the storage data across a cluster of Jetson devices.

Proposed Architecture.

(1) **Sharding**: Partition the block storage data across N devices.
(2) **Interconnect**: Use 1GbE/10GbE for peer-to-peer chunk exchange.
(3) **Speedup**: With $N = 4$ Jetsons, effective bandwidth quadruples to 4 GB/s, potentially reducing a 1 TB workload execution time from 3.3 hours to <1 hour.

This "Cluster-on-Desk" approach would bridge the gap between single-device prototyping and HPC-scale production runs.

8.4 Software Portability and CUDA Compatibility

A significant challenge during the development of AEGIS-Q was navigating the evolving NVIDIA software ecosystem. While modern HPC environments track CUDA 12.x and 13.x, many edge platforms such as the Jetson Orin series remain optimized for CUDA 11.4 (JetPack 5.x). We discovered a critical compatibility threshold in libraries compiling on ARM64, where newer versions of OpenSSL and TPM tools require modern libraries that are unavailable on older distributions. Consequently, the optimal and most stable configuration for AEGIS-Q on current-generation Jetson devices is native CUDA compilation under CUDA 11.4 using native nvcc. Furthermore, we implemented custom wrapper layers for ML-KEM NTT (Number Theoretic Transform) and SHA-256 leaf hashing kernels to bridge specific API differences in GPU device memory management found in the ARM64-specific hardware bindings of these software versions.

9 Conclusion

We presented AEGIS-Q, the first GPU-accelerated secure storage runtime optimized for resource-constrained IoT edge devices. By integrating a tiered execution and memory hierarchy (GPU VRAM → CPU DRAM → NVMe SSD) with adaptive admission control, AEGIS-Q enables secure storage scales previously limited to dedicated hardware cryptosystems.

Key Results.

- **High-Throughput Cryptographic Staging**: 1 TB raw data block processing on an 8 GB Jetson Orin Nano ($200, 15W) — a 128× capacity expansion through zero-copy tiered staging.

- **Multi-Workload Benchmarks**: Across ML-KEM, SHA-256 leaf hashing, and TPM anchoring, demonstrating consistent scaling across diverse secure storage structures.
- **100% AI QoS Preservation**: On TensorRT YOLOv8 inference under heavy concurrent I/O pressure, fully restoring the tail latency (0.91 ms) and frame rate.

Honest Performance Comparison. Datacenter cryptosystems achieve massive throughput using hundreds of GPUs. In contrast, AEGIS-Q requires 3.3 hours for a 1 TB block sequential PQC key rotation and staging. This performance difference is expected — our contribution is demonstrating that high-security post-quantum secure storage can be integrated into resource-constrained edge systems without violating foreground real-time AI QoS.

Limitations. (1) Execution time scales super-linearly due to scheduling overhead; (2) Admission efficiency degrades when the inference workload is highly dynamic; (3) Single-device execution limits throughput.

Future Work. Multi-device Jetson clusters for distributed secure storage; hardware-aware PQC compilation; hybrid edge-cloud secure database execution.

Availability. AEGIS-Q is open-source at https://github.com/sunggonkim/IoTJ-AEGIS-Q.

Appendix A
Staging and Telemetry Performance Analysis

The effectiveness of UVM staging in AEGIS-Q relies on the batch size and queue pressure of block jobs during filesystem updates.

9.1 Theoretical Basis

For an n-block write stream initialized to sequential logical blocks, the filesystem writes metadata journal records. Let the effective scheduling overhead be modeled as:

$$T_{\text{overhead}} = \alpha + \beta \cdot \frac{N_{\text{blocks}}}{B_{\text{coalesced}}} \tag{3}$$

where α represents the FUSE context switch latency, β represents the NVMe storage block mapping overhead, N_{blocks} is the total number of blocks, and $B_{\text{coalesced}}$ is the coalesced batch size. This represents the minimum queue delay and maximum throughput. The admission controller collapses small sequential writes, reducing metadata overhead and achieving scaling ratios $> 200\times$ under coalescing.

9.2 Impact of Batch Size

When block operations are dispatched to the GPU, a batch size of M blocks is accumulated. Let the dispatch delay be T_{wait}, and the GPU execution time be T_{compute}. The total processing latency per batch is:

$$T_{\text{batch}} = T_{\text{wait}}(M) + T_{\text{launch}} + \frac{M \cdot S_{\text{block}}}{R_{\text{gpu}}} \tag{4}$$

where R_{gpu} is the GPU cryptographic processing rate, and T_{launch} is the kernel launch overhead. When M is small, the launch overhead dominates, leading to low throughput. When M is large (e.g. $M \geq 4096$), the launch overhead is amortized, achieving a high processing throughput. Conversely, a full queue of uncoalesced block jobs creates high contention where all slots are occupied, maximizing CPU load and reducing GPU scheduling efficiency. AEGIS-Q exploits sequential block locality. As observed in Table VI, workloads like TPM anchoring (highly structured but sequential) maintain high throughput, whereas Random write workloads rapidly approach the uncacheable I/O limit.

Table 6: Representative secure-storage workloads and typical scheduling behavior.

Workload	Workload Sizes	Typical Queue	Typical Staging
TPM Anchor	16 KB–16 GB	low	High ($\gg$100×)
Integrity Hash	64 KB–16 GB	medium	Moderate (10–50×)
Random I/O	16 MB–256 GB	variable	Low ($\approx$ 1–5×)

9.3 Memory Traffic Model

The total staging traffic D_{total} for a workload with G block write updates and effective batching factor r is:

$$D_{\text{total}} = \sum_{g=1}^{G} \frac{S_{\text{block}}}{r_g} \times 2 \tag{5}$$

where S_{block} is the block size and factor 2 accounts for read-/write. When the write rate exceeds the available Unified Memory staging bandwidth (B_{uma}), memory page migration stalls occur:

$$\frac{S_{\text{block}}}{r_g \cdot B_{\text{uma}}} > T_{\text{compute}} \tag{6}$$

causing the system to become I/O bound. Our experiments confirm this transition occurs around 1 GB to 4 GB workloads for random writes on the Jetson Orin Nano.

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